

Electron configuration and shells

Science understanding

 Visual/Spatial  Logical/Mathematical

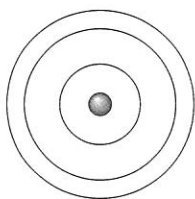
Electrons are arranged around the nucleus of an atom in shells. Electrons fill the shell closest to the nucleus first, then the second shell, then the third shell and so on.

A maximum of 2 electrons can be in the first shell ($n = 1$), 8 in the second shell ($n = 2$), 18 in the third shell ($n = 3$) and 32 in the fourth shell ($n = 4$). The number of electrons that a shell can hold is calculated using the formula $2n^2$, where n is the number of the electron shell.

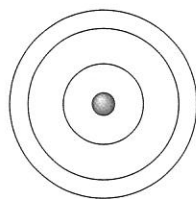
Use this information to **calculate** the electron configuration of the first 18 elements of the periodic table. Complete the diagrams to show how the electrons are arranged. Sodium (Na) has already been done for you (number 11).

configuration (n)
arrangement;
position

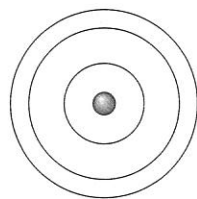
1 H _____



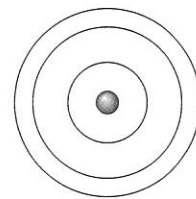
2 He _____



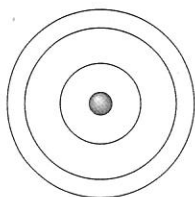
3 Li _____



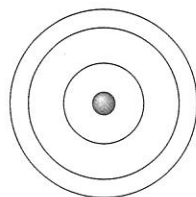
4 Be _____



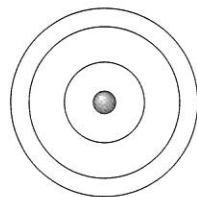
5 B _____



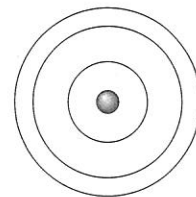
6 C _____



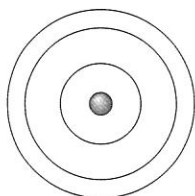
7 N _____



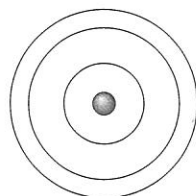
8 O _____



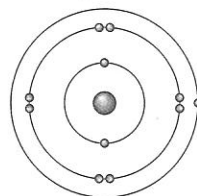
9 F _____



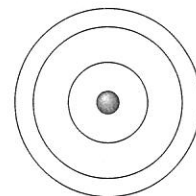
10 Ne _____



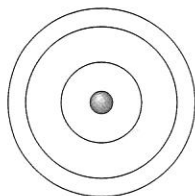
11 Na 2,8,1



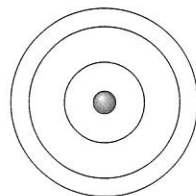
12 Mg _____



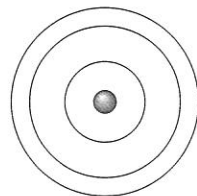
13 Al _____



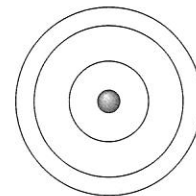
14 Si _____



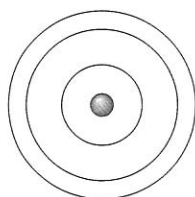
15 P _____



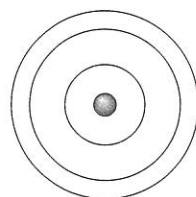
16 S _____



17 Cl _____



18 Ar _____



HINT

The number of each element is its atomic number. This also tells you the number of electrons.

Science inquiry

 Visual/Spatial
  Logical/Mathematical

This table shows the melting points, boiling points and atomic radii of the first 20 elements of the periodic table.

radii (n) plural of radius

Element	Atomic number	Melting point (°C)	Boiling point (°C)	Atomic radius (Å)
Hydrogen, H	1	-259	-253	2.1
Helium, He	2	-272	-269	1.9
Lithium, Li	3	181	1347	1.6
Beryllium, Be	4	1278	2970	1.1
Boron, B	5	2300	2550	1.0
Carbon, C	6	3500	4827	0.9
Nitrogen, N	7	-210	-196	0.9
Oxygen, O	8	-218	-183	0.7
Fluorine, F	9	-220	-188	0.6
Neon, Ne	10	-249	-246	0.5
Sodium, Na	11	98	883	1.9
Magnesium, Mg	12	650	1107	1.6
Aluminium, Al	13	660	2467	1.4
Silicon, Si	14	1410	2355	1.3
Phosphorus, P	15	44	280	1.3
Sulfur, S	16	113	445	1.3
Chlorine, Cl	17	-101	-35	1.0
Argon, Ar	18	-189	-186	0.9
Potassium, K	19	64	774	2.4
Calcium, Ca	20	839	1487	2.0

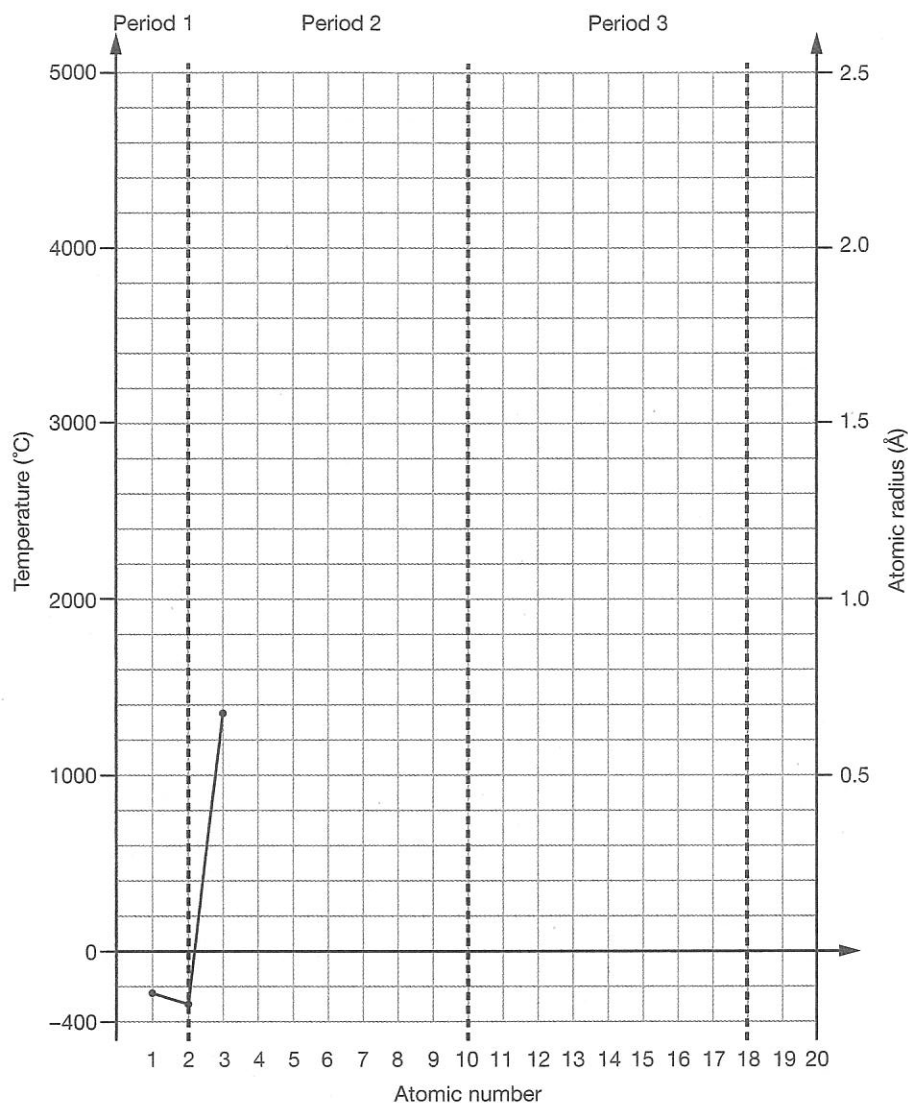
- 1 Use the grid on the next page to **construct** a line graph showing how boiling point varies with atomic number. Use the vertical axis on the left. The graph has been started for you.
- 2 On the same grid, **construct** another line graph showing how melting point varies with atomic number. Use a different colour.
- 3 **Construct** another line graph showing how the atomic radius varies with atomic number. Use the vertical axis on the right of the grid.
- 4 **Compare** the shape of the melting and boiling point graphs.

vertical (adj) in an up and down direction

- 5 **Describe** the overall trend in the melting point, boiling point and atomic radius as you move across a period of the periodic table.

trend (n) direction; pattern

As you move across the period of the periodic table, the melting point _____, the boiling point _____ and the atomic radius _____.



- 6 **Describe** what happens to the trend in the melting point, boiling point and atomic radius when you move from one period to another.

- 7 **Use** your graph to **determine** what happens to the size of an atom as you move down a group of the periodic table.

As you move down a group, the atoms _____

- 8 **Use** the shell structure to **explain** your observations from question 7.

HINT

Look back at worksheet 4.1 to see the shell structure.

Science as a human endeavour

Verbal/Linguistic

Many scientists contributed to our understanding of the elements, their properties and the family relationships between them. Each of these scientists therefore contributed to the development of the periodic table we use today. Some of these scientists are discussed in the Science as a Human Endeavour on pages 115–16 of your student book. This table provides some more information about them.

Scientist	Lived	Scientific contributions	Other facts
Antoine-Laurent de Lavoisier	1743–93	• Did not discover any new elements • Identified and named the existing 33 elements • Proved that mass was not destroyed in a reaction • Proved that rusting involved oxygen bonding to iron (increasing its mass) • Proved that combustion needed oxygen	• Nationality: French • Was very wealthy; worked as a tax collector • Had 13 000 beakers in his laboratory • Was executed by guillotine after the French Revolution • His statue in Paris was made with a 'spare' head from another statue
John Dalton	1766–1844	• Revived the Ancient Greek idea of atoms • Gave each element a pictorial symbol, which is not used today • Predicted atomic masses • Developed laws that explain how gases behave • Showed that compounds were made of elements in definite mass proportions	• Nationality: English • Came from a very poor family • Was put in charge of his small school when he was 12 years old • Still taught mathematics at a primary school when he was old and famous
Jons Jacob Berzelius	1779–1848	• Used atomic mass to arrange the elements • Gave the elements symbols that we still use today, based on their Greek or Latin names • Showed that compounds were made of elements in definite mass proportions • Was the first to classify substances as organic (containing C) or inorganic (no C)	• Nationality: Swedish • Used superscript numbers instead of subscript numbers in chemical formulas—wrote water as H^2O instead of H_2O
Johann Dobereiner	1780–1849	• Placed elements into 'triads' (families of three)	• Nationality: German



Scientist	Lived	Scientific contributions	Other facts
John Newlands	1837–98	- Used atomic mass to arrange elements - Noticed every eighth element 'harmonising' (forming patterns) - Placed elements in periods of 7 elements	- Nationality: English - Called his arrangement 'the law of octaves' after musical octaves
Lothar Meyer	1830–95	- Constructed a table very similar to Mendeleev's - Finished his periodic table before Mendeleev - Published his table after Mendeleev	- Nationality: German - Is regarded as a 'father' of the periodic table
Dmitri Ivanovich Mendeleev	1834–1907	- Used atomic mass to arrange elements - Like Newlands, arranged the elements in periods of 7 - Arranged related elements in columns like Dobereiner's triads - Left gaps for undiscovered elements (e.g. Ge)	- Nationality: Russian - Refused to accept that electrons and radiation and radioactivity existed - Has a radioactive element named after him—mendelevium!
William Ramsay	1852–1916	- Discovered the noble, or inert, gases - Set them as a separate group	- Nationality: Scottish - Received the Nobel Prize in Chemistry
Henry Moseley	1887–1915	- Used atomic number, not mass, to arrange elements - Produced the periodic table still in use today	- Nationality: English - Was shot through the head and killed in World War I at Gallipoli

harmonise (v) to be similar in some way

octave (n) a group of eight notes in music

Use the information in the table and information from Unit 4.2 in your student book to **identify**:

(a) the scientist who was Scottish _____

(b) the scientist who finished his table before Mendeleev did but received less recognition _____

(c) the scientist who thought that the elements harmonised in octaves like music _____

(d) the scientist who received a Nobel Prize _____

- (e) the scientist who wrote water as H^2O _____
- (f) the scientist who left gaps in his table _____
- (g) the scientist who developed triads _____
- (h) the scientist who thought that light was an element _____
- (i) the scientist who is considered to be the 'father' of the modern periodic table

- (j) the scientist who developed gas laws _____
- (k) the scientist who arranged elements according to their atomic number and not their mass _____
- (l) the scientist who was shot in the head _____
- (m) the scientist who had his head chopped off _____
- (n) the scientist who was Russian _____
- (o) the scientist who revived the idea that atoms existed _____
- (p) the scientist who had 13 000 beakers in his laboratory _____
- (q) the scientist who was put in charge of his school when he was 12

- (r) the scientist who grouped chemicals as organic or inorganic _____
- (s) the scientist who gave the elements the symbols we use today _____
- (t) the scientist who has a radioactive element named after him, even though he didn't believe in radioactivity _____
- (u) the scientist who proved that iron got heavier when it rusted _____
- (v) the scientist who had a statue that should have had its head chopped off because it was the wrong one! _____

Families of the periodic table

Science understanding

Visual/Spatial

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

20

Ca

calcium

21

Sc

scandium

22

Ti

titanium

23

V

vanadium

24

Cr

chromium

25

Mn

manganese

26

Fe

iron

27

Co

cobalt

28

Ni

nickel

29

Cu

copper

30

Zn

zinc

31

Ga

gallium

32

Ge

germanium

33

As

arsenic

34

Se

selenium

35

Br

bromine

36

Kr

krypton

37

Rb

rubidium

38

Sr

strontium

39

Y

yttrium

40

Zr

zirconium

41

Nb

niobium

42

Mo

molybdenum

43

Tc

technetium

44

Ru

ruthenium

45

Rh

rhodium

46

Pd

palladium

47

Ag

silver

48

Cd

cadmium

49

In

indium

50

Sn

tin

51

Sb

antimony

52

Te

tellurium

53

I

iodine

54

Xe

xenon

55

Cs

caesium

56

Ba

barium

57–71

lanthanides

72

Hf

hafnium

73

Ta

tantalum

74

W

tungsten

75

Re

rhenium

76

Os

osmium

77

Ir

iridium

78

Pt

platinum

79

Au

gold

80

Hg

mercury

81

Tl

thallium

82

Pb

lead

83

Bi

bismuth

84

Po

polonium

85

At

astatine

86

Rn

radon

87

Fr

franchium

88

Ra

radium

89–103

actinides

104

Rf

rutherfordium

105

Db

dubnium

106

Sg

seaborgium

107

Bh

bohrium

108

Hs

hassium

109

Mt

meitnerium

110

Ds

darmstadtium

111

Rg

roentgenium

112

Cn

copernicium

113

Uut

ununtrium

114

Uuq

ununquadium

115

Uup

ununpentium

116

Uuh

ununhexium

117

Uus

ununseptium

118

Uuo

ununoctium

13

Al

aluminium

5

B

boron

6

C

carbon

7

N

nitrogen

8

O

oxygen

9

F

fluorine

10

Ne

neon

13

Al

aluminium

14

Si

silicon

15

P

phosphorus

16

S

sulfur

17

Cl

chlorine

18

Ar

argon

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

20

Ca

calcium

21

Sc

scandium

22

Ti

titanium

23

V

vanadium

24

Cr

chromium

25

Mn

manganese

26

Fe

iron

27

Co

cobalt

28

Ni

nickel

29

Cu

copper

30

Zn

zinc

31

Ga

gallium

32

Ge

germanium

33

As

arsenic

34

Se

selenium

35

Br

bromine

36

Kr

krypton

37

Rb

rubidium

38

Sr

strontium

39

Y

yttrium

40

Zr

zirconium

41

Nb

niobium

42

Mo

molybdenum

43

Tc

technetium

44

Ru

ruthenium

45

Rh

rhodium

46

Pd

palladium

47

Ag

silver

48

Cd

cadmium

49

In

indium

50

Sn

tin

51

Sb

antimony

52

Te

tellurium

53

I

iodine

54

Xe

xenon

55

Cs

caesium

56

Ba

barium

57–71

lanthanides

72

Hf

hafnium

73

Ta

tantalum

74

W

tungsten

75

Re

rhenium

76

Os

osmium

77

Ir

iridium

78

Pt

platinum

79

Au

gold

80

Hg

mercury

81

Tl

thallium

82

Pb

lead

83

Bi

bismuth

84

Po

polonium

85

At

astatine

86

Rn

radon

87

Fr

franchium

88

Ra

radium

89–103

actinides

104

Rf

rutherfordium

105

Db

dubnium

106

Sg

seaborgium

107

Bh

bohrium

108

Hs

hassium

109

Mt

meitnerium

110

Ds

darmstadtium

111

Rg

roentgenium

112

Cn

copernicium

113

Uut

ununtrium

114

Uuq

ununquadium

115

Uup

ununpentium

116

Uuh

ununhexium

117

Uus

ununseptium

118

Uuo

ununoctium

57

La

lanthanum

58

Ce

cerium

59

Pr

praseodymium

60

Nd

neodymium

61

Pm

promethium

62

Sm

samarium

63

Eu

europium

64

Gd

gadolinium

65

Tb

terbium

66

Dy

dysprosium

67

Ho

holmium

68

Er

erbium

69

Tm

thulium

70

Yb

ytterbium

71

Lu

lutetium

89

Ac

actinium

90

Th

thorium

91

Pa

protactinium

92

U

uranium

93

Np

neptunium

94

Pu

plutonium

95

Am

americum

96

Cm

curium

97

Bk

berkelium

98

Cf

californium

99

Es

einsteinium

100

Fm

fermium

101

Md

mendelevium

102

No

nobelium

103

Lr

lawrencium

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

20

Ca

calcium

21

Sc

scandium

22

Ti

titanium

23

V

vanadium

24

Cr

chromium

25

Mn

manganese

26

Fe

iron

27

Co

cobalt

28

Ni

nickel

29

Cu

copper

30

Zn

zinc

31

Ga

gallium

32

Ge

germanium

33

As

arsenic

34

Se

selenium

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hafnium

73

Ta

tantalum

74

W

tungsten

75

Re

rhenium

76

Os

osmium

77

Ir

iridium

78

Pt

platinum

79

Au

gold

80

Hg

mercury

81

Tl

thallium

82

Pb

lead

83

Bi

bismuth

84

Po

polonium

85

At

astatine

86

Rn

radon

87

Fr

franchium

88

Ra

radium

89–103

actinides

104

Rf

rutherfordium

105

Db

dubnium

106

Sg

seaborgium

107

Bh

bohrium

108

Hs

hassium

109

Mt

meitnerium

110

Ds

darmstadtium

111

Rg

roentgenium

112

Cn

copernicium

113

Uut

ununtrium

114

Uuq

ununquadium

115

Uup

ununpentium

116

Uuh

ununhexium

117

Uus

ununseptium

118

Uuo

ununoctium

57

La

lanthanum

58

Ce

cerium

59

Pr

praseodymium

60

Nd

neodymium

61

Pm

promethium

62

Sm

samarium

63

Eu

europium

64

Gd

gadolinium

65

Tb

terbium

66

Dy

dysprosium

67

Ho

holmium

68

Er

erbium

69

Tm

thulium

70

Yb

ytterbium

71

Lu

lutetium

89

Ac

actinium

90

Th

thorium

91

Pa

protactinium

92

U

uranium

93

Np

neptunium

94

Pu

plutonium

95

Am

americum

96

Cm

curium

97

Bk

berkelium

98

Cf

californium

99

Es

einsteinium

100

Fm

fermium

101

Md

mendelevium

102

No

nobelium

103

Lr

lawrencium

1

H

hydrogen

3

Li

lithium

11

Na

sodium

4

Be

beryllium

12

Mg

magnesium

19

K

potassium

20

Ca

calcium

21

Sc

scandium

22

Ti

titanium

23

V

vanadium

24

Cr

chromium

25

Mn

manganese

26

Fe

iron

27

Co

cobalt

28

Ni

nickel

29

Cu

copper

30

Zn

zinc

31

Ga

gallium

32

Ge

germanium

33

As

arsenic

34

Se

selenium

35

Br

bromine

36

Kr

krypton

37

Rb

rubidium

38

Sr

strontium

39

Y

yttrium

40

Zr

zirconium

41

Nb

niobium

42

Mo

molybdenum

43

Tc

technetium

44

Ru

ruthenium

45

Rh

rhodium

46

Pd

palladium

47

Ag

silver

48

Cd

cadmium

49

In

indium

50

Sn

tin

51

Sb

antimony

52

Te

tellurium

53

I

iodine

54

Xe

xenon

55

Cs

caesium

56

Ba

barium

57–71

lanthanides

72

Hf

hafnium

73

Ta

tantalum

74

W

tungsten

75

Re

rhenium

76

Os

osmium

77

Ir

iridium

78

Pt

platinum

79

Au

gold

80

Hg

mercury

81

Tl

thallium

82

Pb

lead

83

Bi

bismuth

84

Po

polonium

85

At

astatine

86

Rn

radon

87

Fr

franchium

88

Ra

radium

89–103

actinides

104

Rf

rutherfordium

105

Db

dubnium

106

Sg

seaborgium

107

Bh

bohrium

108

Hs

hassium

109

Mt

meitnerium

110

Ds

darmstadtium

111

Rg

roentgenium

112

Cn

copernicium

113

Uut

ununtrium

114

Uuq

ununquadium

115

Uup

ununpentium

116

Uuh

ununhexium

117

Uus

ununseptium

118

Uuo

ununoctium

57

La

lanthanum

58

Ce

cerium

59

Pr

praseodymium

60

Nd

neodymium

61

Pm

promethium

62

Sm

samarium

63

Eu

europium

64

Gd

gadolinium

65

Tb

terbium

66

Dy

dysprosium

67

Ho

holmium

68

Er

erbium

69

Tm

thulium

70

Yb

ytterbium

1 Use different colours or shading to **demonstrate** where the following elements are on the simple version of the periodic table shown above. Note that some of your coloured or shaded areas might overlap.

- noble or inert gases
- alkali metals
- actinides
- transition elements
- halogens
- alkaline earths
- lanthanides

overlap (v) for one thing to partly cover another

2 Circle an element symbol to **identify**:

- the only non-metal that exists as a liquid at normal room temperature
- the element found in all living (and dead) things, being the basis for most of their molecules
- the lightest of all elements
- the heaviest of all elements.

Science inquiry

 Verbal/Linguistic  Visual/Spatial

- 1 A carbon atom has four outer-shell electrons, placing it in group 14 of the periodic table. When it bonds, these four electrons are shared with other carbon atoms or atoms of other non-metals to form four covalent bonds. This ability to form four bonds gives carbon the capacity to form some amazing molecules. Carbon-based molecules range from small groupings of atoms to long chains and rings that have carbon atoms as their backbone. Some are shown in Figure 4.5.1.

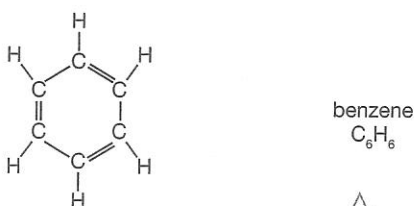
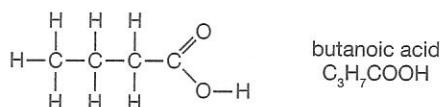
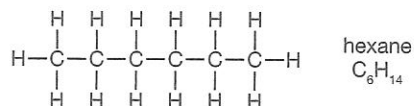


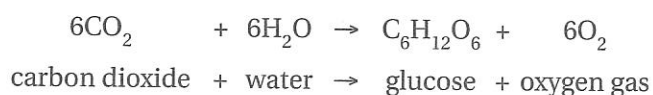
Figure
4.5.1

- 2 These carbon backbones give the molecules a new classification too. They are classified as *organic*. Organic chemistry is the study of these molecules, their properties and their reactions.

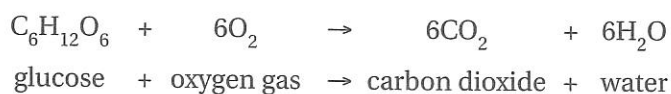
- 3 Organic molecules form the basis of life on Earth. They make up:

- the living tissue of animals, such as their skin and organs
- the leaves, stems, trunks and roots of plants
- the cell walls of bacteria and other micro-organisms.

- 4 Carbon becomes part of plants via photosynthesis, in the following reaction:



- 5 During photosynthesis plants take in carbon dioxide, break it up and construct a new organic molecule called glucose ($C_6H_{12}O_6$). Some of this is used immediately in respiration to give the energy the plant needs. The respiration reaction is:

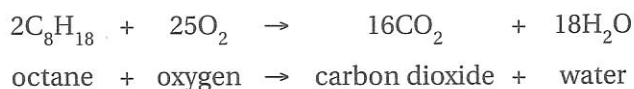


- 6 The rest of the glucose is stored within the plants in the form of other organic molecules such as sucrose ($C_{12}H_{22}O_{11}$) and starch. Animals eat plants or eat other animals that eat plants. In this way, the carbon stored in the plant becomes part of the animals' structures too.
- 7 Dead organisms also contain organic molecules. For this reason, substances such as wood and paper (from plants) and leather (from animals) are organic. Fossil fuels such as oil, petrol, coal and gas are organic too because they are made from the remains of long-dead organisms. Plastics are made from oil and so they are organic as well.
- 8 When burnt, wood, paper, plastics and fossil fuels release much of their carbon into the atmosphere as carbon dioxide (CO_2). For example, petrol contains octane (C_8H_{18}).

LB 4.5

p. 138

Its combustion reaction is:



- 9 Carbon dioxide is a known greenhouse gas. Scientists strongly suspect that its increasing concentration in the atmosphere is causing global warming. Average global temperatures are rising as a result, leading to a change in the climate.

- 1 For the molecules in Figure 4.5.1, **state** how many bonds each carbon atom has.

- 2 Vegetables, meat and other products are often described as 'organic' if they are pesticide- and herbicide-free and have only been fertilised with natural compost and manure. **Propose** reasons why the term *organic* is used this way when it has such a different scientific meaning. (paragraph 3)

HINT

Are all pesticides and herbicides organic?

- 3 **Propose** what the following scientists study.

(a) An organic chemist studies _____

(b) An inorganic chemist studies _____

- 4 **Compare** the structures of hexane and benzene by listing their similarities and differences.

Similarities: _____

Differences: _____

- 5 **Explain** where the carbon in the following comes from.

(a) plants (paragraphs 4 and 5) _____

(b) animals (paragraph 6) _____

(c) fossil fuels (paragraph 7) _____

(d) plastics (paragraph 7) _____

- 6 **Explain** why oil, petrol, gas and coal are known as *fossil fuels*. (paragraph 7)

- 7 **Explain** why many scientists link climate change to the combustion of fossil fuels. (paragraphs 8 and 9)

Science inquiry

 Visual/Spatial
  Logical/Mathematical

systematic (adj)
ordered; logical

There are many thousands of different organic molecules. To avoid confusion, they must be named in a systematic way. The International Union of Pure and Applied Chemistry (IUPAC) has developed rules for naming organic molecules.

The simplest of all organic molecules are the hydrocarbons. These molecules are made of just carbon and hydrogen atoms. Depending on their bonding, hydrocarbons can be classified as alkanes, alkenes or alkynes. The first ten of each family are shown below.

Hydrocarbon family

	Alkanes	Alkenes	Alkynes
General formula	C_nH_{2n+2}	C_nH_{2n}	C_nH_{2n-2}
$n = 1$	Methane, CH_4 $\begin{array}{c} H \\ \\ H-C-H \\ \\ H \end{array}$	None exists	None exists
$n = 2$	Ethane, C_2H_6 $\begin{array}{cc} H & H \\ & \\ H-C & -C-H \\ & \\ H & H \end{array}$	Ethene, C_2H_4 $\begin{array}{cc} H & & H \\ & \backslash & / \\ & C=C \\ & / & \backslash \\ H & & H \end{array}$	Ethyne, C_2H_2 $H-C \equiv C-H$
$n = 3$	Propane, C_3H_8 $\begin{array}{ccc} H & H & H \\ & & \\ H-C & -C & -C-H \\ & & \\ H & H & H \end{array}$	Propene, C_3H_6 $\begin{array}{ccc} H & & H \\ & \backslash & / \\ & C=C & -C-H \\ & / & \\ H & & H \end{array}$	Propyne, C_3H_4 $H-C \equiv C-\begin{array}{c} H \\ \\ C-H \\ \\ H \end{array}$
$n = 4$	Butane, C_4H_{10}	Butene, C_4H_8	Butyne, C_4H_6
$n = 5$	Pentane, C_5H_{12}	Pentene, C_5H_{10}	Pentyne, C_5H_8
$n = 6$	Hexane, C_6H_{14}	Hexene, C_6H_{12}	Hexyne, C_6H_{10}
$n = 7$	Heptane, C_7H_{16}	Heptene, C_7H_{14}	Heptyne, C_7H_{12}
$n = 8$	Octane, C_8H_{18}	Octene, C_8H_{16}	Octyne, C_8H_{14}
$n = 9$	Nonane, C_9H_{20}	Nonene, C_9H_{18}	Nonyne, C_9H_{16}
$n = 10$	Decane, $C_{10}H_{22}$	Decene, $C_{10}H_{20}$	Decyne, $C_{10}H_{18}$

1 Name the organisation that determines the names of different organic compounds.

- 2 Different scientists might speak different languages but they all use the same names for chemicals. **Propose** what science would be like if they didn't.

- 3 **Identify** which family of hydrocarbons (alkanes, alkenes or alkynes) has:

(a) only single bonds _____

(b) a double bond _____

(c) a triple bond _____

- 4 **Compare** the alkanes, alkenes and alkynes by listing their similarities and differences.

Similarities: _____

Differences: _____

- 5 **Use** the general formulas given in the table on the previous page to **calculate** the chemical formulas of the:

$$n = 12 \text{ so } 2n + 2 = 26$$

(a) 12th member of the alkanes ($n = 12$) *so formula is $C_{12}H_{26}$* _____

(b) 20th member of the alkenes ($n = 20$) _____

(c) 30th member of the alkynes ($n = 30$) _____

- 6 To get the next member of the alkane family, you add a grouping of carbon and hydrogen atoms. **Identify** which of the following groupings needs to be added to get the next alkane: CH , CH_2 , CH_4 , CH_6 .

Each alkane increases by 1 carbon atom and 2 hydrogen atoms, so you would need to add _____.

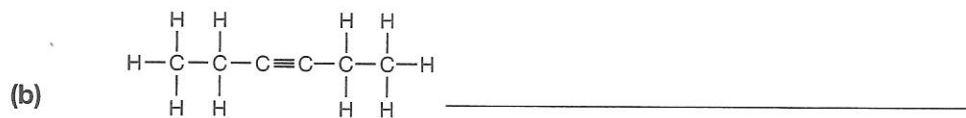
- 7 **Use** the list in question 6 to **identify** the grouping that needs to be added to get the next:

(a) alkene _____

(b) alkyne _____

- 8 **Propose** reasons why there is no alkene or alkyne with just one carbon atom.

9 Propose the names of the following hydrocarbons.



10 Construct diagrams of what you think the following molecules would look like.

(a) butene

(b) pentane

(c) octyne

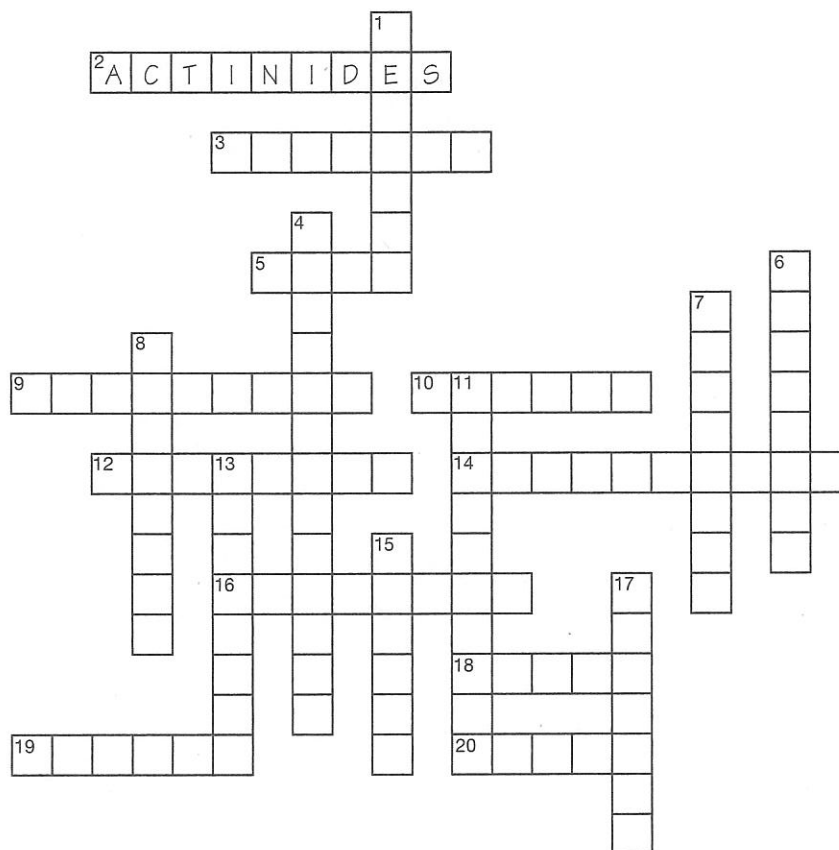
HINT

Refer to the table on page 50 to see how the atoms are arranged.

Science understanding



Recall your knowledge of the periodic table to complete this crossword puzzle.



Across

- 2 One of two special blocks of elements at the bottom of the periodic table
- 3 Extended 3D structure
- 5 Formed when electrons transfer from one atom to another
- 9 Exists as sole atoms
- 10 Number that tells you how many protons are in the nucleus
- 12 Small group of atoms bonded together
- 14 Different forms of the same element
- 16 Bonding based on an electron 'sea'
- 18 Bonding that involves complete transfer of electrons
- 19 Vertical columns in the periodic table
- 20 Group 18 gases

Down

- 1 Horizontal rows in the periodic table
- 4 Shows how electrons are arranged in their shells
- 6 Bonding between non-metallic atoms
- 7 The lightest of all the elements
- 8 Non-metals that are in group 17
- 11 A special block of elements in groups 3–12
- 13 118 of these are known to exist
- 15 Metals that are in group 1
- 17 Core of an atom